

FLOODGAUGE · DESIGN PRECIS

FloodGauge design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices adopted for TRL 3 pending Amish's review (FLG-DDR-001); numbers checked in FLG-CAL-001; parametric model and drawing FLG-DWG-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

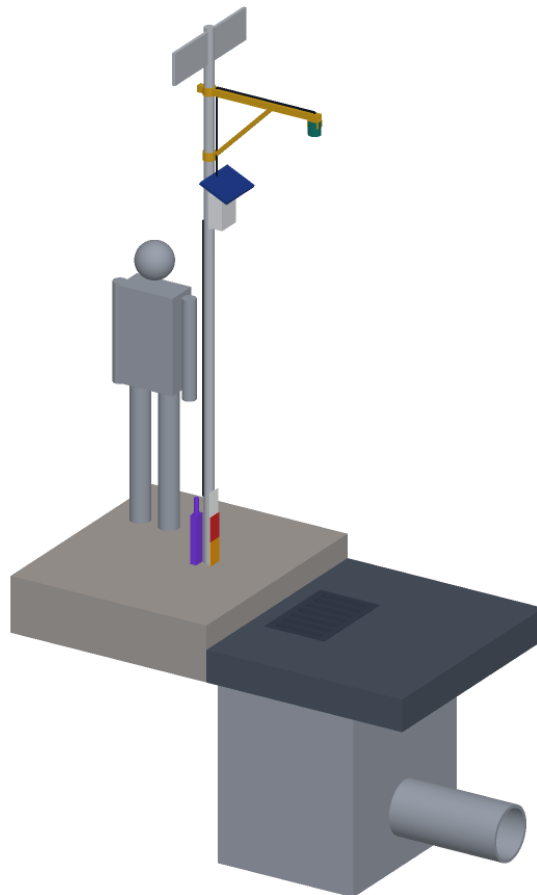
FloodGauge design precis

■ Summary

FloodGauge is a FieldNode core on an existing street pole next to a storm inlet, with two ranging heads: a 60 GHz radar head on a short arm over the gutter that measures water depth on the street, and an ultrasonic head in a slotted stilling tube inside the catch basin that sees the drain filling before the street floods. The node turns ranges into levels, raises its sampling rate when water rises, and sends levels only over LoRaWAN to an alert service that warns crews and residents at default depths of 150 mm and 300 mm. FLG-CAL-001 gives an alert latency of 82 s at the 95th percentile (110 s worst case), a sensor load under 10 mW, 4.62 kg on the pole and \$145.50 of FloodGauge-specific parts against the \$150 budget (\$271.50 with the \$126.00 FieldNode core, which is costed in its own repo). Two requirements are not met: the drain head cannot read the top 290 mm of the basin (R3), and the conduit makes installation civil work (R12). The design choices below are adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review (FLG-DDR-001).

FloodGauge

CONCEPT, NOT FOR FABRICATION



Grey: Person, 1.75 m (scale) for scale

Figure 1. Concept massing model beside a storm inlet, with a 1.75 m person for scale. CONCEPT, NOT FOR FABRICATION.

■ How it works

1. **Range the street.** The street head (item 2) looks straight down at the gutter from 2.95 m, 250 mm past the curb face. Water depth is the surveyed head height minus the measured range. The beam also lights the curb, so a dry-road background is recorded at installation and the firmware tracks the peak directly below the head (FLG-CAL-001, A2).
2. **Range the drain.** The drain head (item 4) sits at the top of a slotted 75 mm tube (item 5) fixed to the basin wall. Water in the tube follows the basin level (lag under 5 mm even at 50 mm/s), and the tube walls give the wide ultrasonic beam a clean path, so the head sees the water surface and not the basin walls. A wet probe on the head signals when the basin is full to the head, and an NTC in the head compensates the speed of sound.
3. **Decide on the node.** FieldNode (item 1), mounted 2.35 m above the road, samples both heads every 60 s. When either level rises faster than 20 mm/min or passes the first band, it samples every 10 s. It applies plausibility checks: a sudden 1 m "rise" on the street head with an empty drain is a parked vehicle or a person, not a flood.
4. **Report.** Normally one uplink every 15 min; in an event an uplink on each band crossing and then each minute. Payloads carry levels, rate of rise, battery and status only.
5. **Alert.** A gateway (TwinKit or a city LoRaWAN network) passes readings to an open alert service that sends crew notifications and opt-in resident messages and publishes open data.
6. **Check by eye.** A depth marker plate (item 8) with bands at 150 and 300 mm above the road lets residents and crews read depth directly and check the gauge.

FloodGauge: cutaway

CONCEPT, NOT FOR FABRICATION

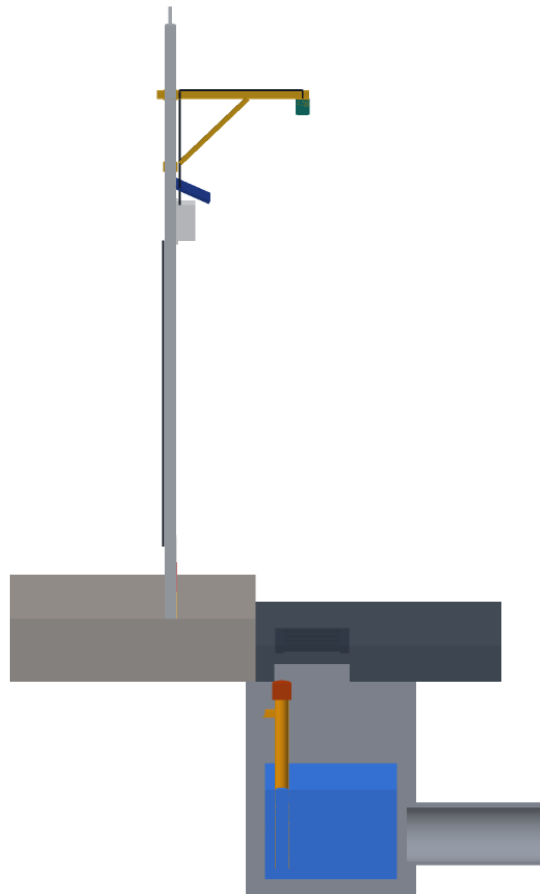


Figure 2. Section across the street through the inlet: street head over the gutter, stilling tube and drain head in the catch basin, conduit under the sidewalk. Water level is illustrative.

FloodGauge: data flow from water surface to alert (values are estimates)

CONCEPT, NOT FOR FABRICATION

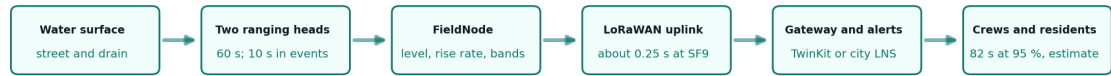


Figure 3. Data flow from the water surface to an alert. Values are estimates or proposals.

■ Main components

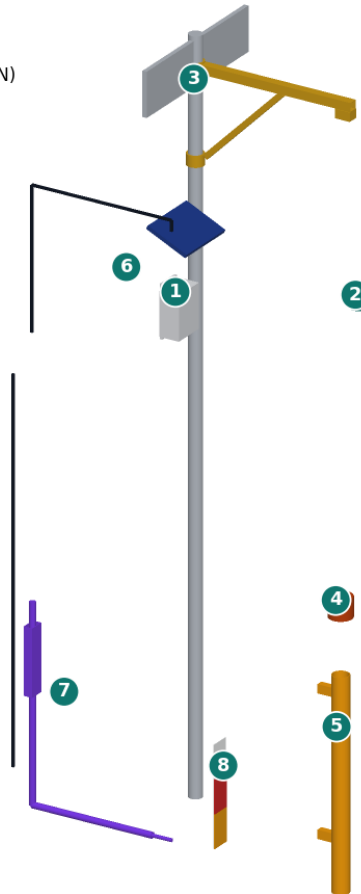
Table 1. Main components (numbers match the exploded view and `bom/bom.csv`)

#	Component	Adopted choice (FLG-DDR-001)	Notes
1	FieldNode core	Lab shared node: IP65 enclosure, 6 W panel as hood, 6 Ah LiFePO4 cell, MPPT board with 3.3, 5 and 12 V switched rails, STM32WL-class LoRaWAN, two M12 sensor ports	Center 2.35 m above the road, facing the street; 2.41 kg and \$126.00 per FND-CAL-001
2	Street radar head	60 GHz pulsed coherent radar module (Acconeer XM125 class) behind a PTFE lens in a sealed IP67 housing	D3; accuracy, current and price not yet checked against a data sheet
3	Sensor arm	40 x 40 x 2 mm aluminium tube 805 mm, 25 mm square knee brace at 45°, two band clamps 400 mm apart	Head 250 mm past the curb face over the gutter (D5); height subject to clearance rules
4	Drain head	Waterproof ultrasonic ranger (A02YYUW: 3 to 450 cm, ±1 cm, IP67) with a two-electrode wet probe and an NTC, potted in a sealed cap	Face 300 mm below the road; the transducer face keeps its IP67 seal, so R9 is at risk
5	Stilling tube	75 mm OD PVC, 940 mm long, 40 slots 5 x 50 mm, two stainless wall brackets, mouth 60 mm above the basin floor	Reachable from the surface with the grate lifted
6	Sensor cables	Two outdoor cables with M12 5-pin plugs to the FieldNode ports	Drip loops at every entry
7	Conduit	25 mm conduit cored through the basin wall and run under the sidewalk to the back of the pole, with a steel riser guard	Civil work; needs the asset owner
8	Depth marker plate	Aluminium plate with amber band (150 to 300 mm) and red band (300 to 450 mm) above road level	Visual check and resident information
9	Alert service	Open software on the gateway or a small server (not modeled)	Part of the software license (MIT)

FloodGauge: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 FieldNode core (enclosure, LiFePO4, LoRaWAN)
- 2 Street radar head, 60 GHz
- 3 Sensor arm and pole clamps
- 4 Drain head, ultrasonic with wet probe
- 5 Stilling tube, 75 mm slotted PVC
- 6 Sensor cables, M12
- 7 Conduit and riser guard
- 8 Depth marker plate



Existing street, catch basin and water not shown. Grey: existing sign pole (not in kit).

Figure 4. Exploded view with BOM numbers. The existing street and basin are not shown.

■ First-order numbers

The TRL 2 estimates were checked in FLG-CAL-001 v0.1; the values below are from that note and its script.

Table 2. Key numbers at TRL 3

Quantity	Value	Basis	Requirement
Street head range	2.95 m to dry road, 2.35 m at 600 mm depth	Lens 2.95 m above the road	R1 met on paper
Street depth error, radar	± 6.3 mm root sum square, ± 11.3 mm worst-case sum	Module ± 2 mm (assumed), datum ± 5 mm, pole expansion ± 1.3 mm, ripple ± 3 mm	R2 met on paper
Street depth error, ultrasonic variant	+224 to -148 mm uncompensated over -20 to 50 °C; ± 19.5 mm compensated	Speed of sound $331.3 + 0.606 T$ m/s	R2 not met by this variant
Drain level error	± 11.0 mm with the NTC, ± 19.7 mm without	± 10 mm module (DFRobot), 950 mm range	R4 met on paper
Drain reading range	50 mm above the floor to 330 mm below the road; wet probe at 310 mm	Head face 300 mm below the road, 30 mm blind zone	R3 not met (top 290 mm)
Energy per sampling cycle	85.1 mJ	Radar 59.4 mJ, ultrasonic 12.0 mJ, rails 90 %, controller 5.8 mJ	
Sensor load	1.42 mW at 60 s; 9.3 mW with 10 s sampling and 1 min uplinks all day	1,440 or 8,640 cycles a day	R8 met: 68 days without sun
Uplink airtime	23.7 s/day normal at SF9; 14.8 s/h in an event	246.8 ms per 20-byte uplink	R7 at risk at SF11 and SF12
Alert latency	82 s at the 95th percentile, 110 s worst case without a lost packet	Simulation of 200,000 events	R6 met on paper
Mass on the pole	4.62 kg	FieldNode 2.41 kg plus arm, head, cables and marker	R15 met on paper
Parts cost	\$145.50 FloodGauge-specific; \$271.50 with FieldNode	bom/bom.csv	R16 met on paper

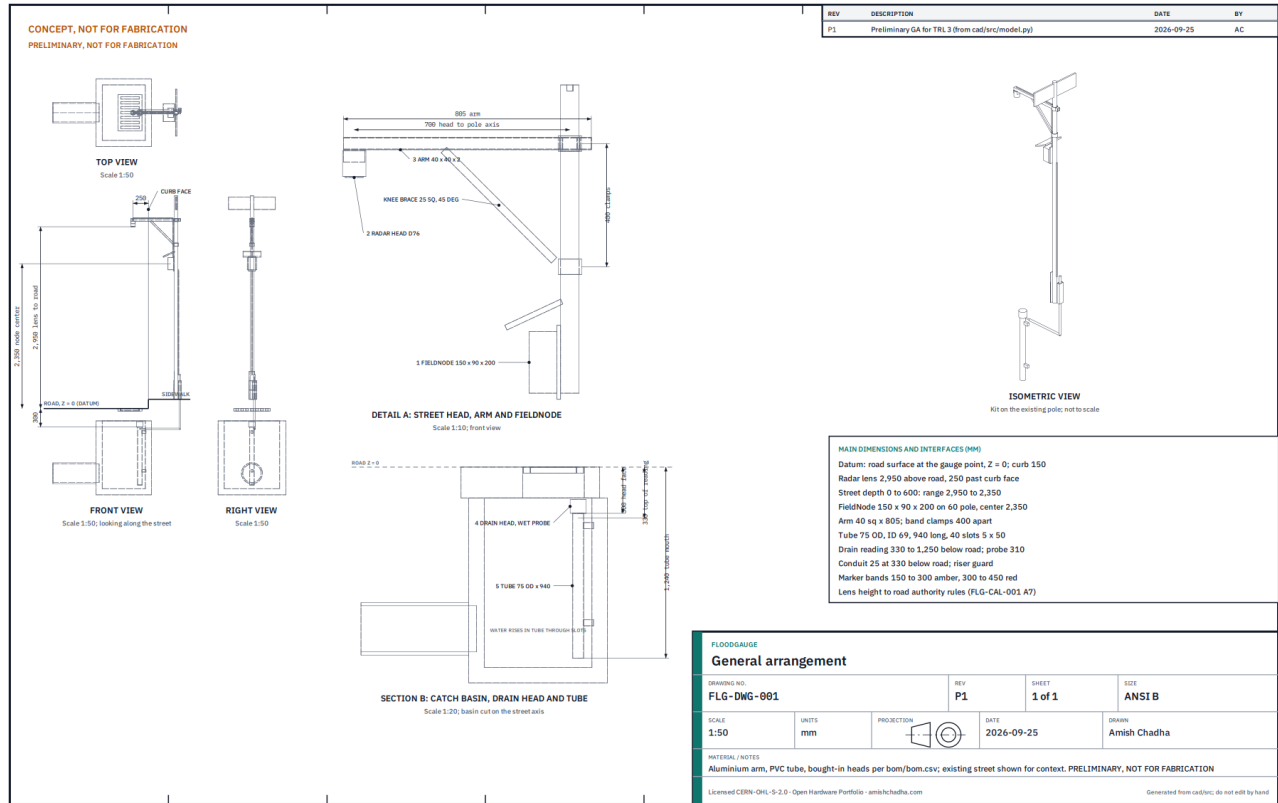


Figure 5. General arrangement FLG-DWG-001, Rev P1, from `cad/src/model.py`. PRELIMINARY, NOT FOR FABRICATION.

■ Key design choices

Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review (FLG-DDR-001, D1 to D9).

- Two heads on one node** (street and drain), because each head misses one of the two ways streets flood (FLG-PRB-001). Street head only remains the fallback where no cable route to the basin exists. (D2)
- Radar for the street head.** FLG-CAL-001 confirms that a compensated ultrasonic head misses R2 (± 19.5 mm). The ultrasonic head stays documented as a lower-cost variant. (D3)
- Ultrasonic in a stilling tube for the drain**, with the wet probe as a second signal and an NTC for temperature. (D4)
- Arm over the gutter** at 2.95 m or higher, subject to local clearance rules. FLG-CAL-001 finds 2.95 m inside a 4.0 to 4.3 m vehicle envelope at the curb; a height of about 4.6 m is proposed where tall vehicles use the curb lane, awaiting Amish. (D5)
- Alert bands at 150 mm and 300 mm** above the road (NWS), plus a drain-full alert from the wet probe, as defaults to be set with the partner city. In a fast flood the 150 mm band warns at about 190 to 250 mm (FLG-CAL-001, F3). (D6)
- Private or city LoRaWAN gateway for event reporting** (TwinKit or the city network), with The Things Network for pilots only. (D7)
- Levels only, no imaging.** (D8)

The FieldNode core is costed in its own repo, and R16 covers FloodGauge-specific parts only (D1). The pitch and problem lines are unchanged (D9).

■ Safety

Safety: Work beside live traffic. Install only with the road and pole owner's permission, with traffic management and high-visibility clothing as local rules require.

Safety: Catch basins are confined spaces that can hold toxic or oxygen-poor air and fast-rising water. The design is installed from the surface; never enter a basin. Lifting a grate can crush fingers and strain backs; use a grate hook and two people. Stormwater carries sewage and chemicals: wear gloves and eye protection and wash after work.

Safety: Work at height on the pole needs a stable ladder or platform, a second person and clearance from overhead power lines.

Safety: An arm over the gutter at 2.95 m can be struck by trucks, buses or their mirrors at the curb (FLG-CAL-001, A7). Mount it at the height the road authority requires, and fit an anti-rotation stop so that wind cannot swing it (twist margin 1.45 on friction alone).

Safety: FieldNode holds a LiFePO4 cell of about 19 Wh. Fuse it, charge only between 0 and 45 °C, and never install a swollen, damaged or wet cell.

Safety: FloodGauge supplements official warnings and must never be presented as the only warning. A dead battery, a blocked tube or a lost radio link can hide a flood. The alert service must report when a gauge goes silent.

■ Open questions

- [] Radar module accuracy, current draw, beam width with the lens, price and radio certification for fixed outdoor use at 60 GHz in the pilot country (assumed in FLG-CAL-001).
- [] Pilot partner and city (FLG-DDR-001, O1).
- [] How to route the drain head cable to the pole without trenching, or whether a second FieldNode in the basin (with its radio underground) is better.
- [] Silt and debris in the stilling tube: slot size, cleaning interval, and whether the tube clogs in the first storm.
- [] Plausibility logic for vehicles, people, snow and floating debris under the street head.
- [] Theft and vandalism protection at 2.5 m.
- [] Alert governance: who receives alerts, who acts, and how residents opt in and out.
- [] Datum survey method and how often to recheck it.